

# **K-Theory Localizations and the Prime Decomposition of Gauge Couplings**

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## Abstract

We develop a K-theoretic refinement of the dimensional-closure mechanism in the World-Layer Model (WLM). While previous work derived the electromagnetic residue  $137$  from a triple congruence system in cohomology, we show that the deeper origin of such residues lies in the prime-localized structure of topological K-theory and KO-theory.

We prove that gauge couplings arise as **finite prime localizations** of a single **global KO-theoretic gravitational class**, and that residues such as  $137$  correspond to **topologically irreducible prime blocks** that survive dimensional projection.

This establishes a unified framework in which:

- gravity = global KO-class (all primes active)
- gauge forces = finite prime localizations
- coupling constants = residues of localized indices

and explains why physically relevant couplings tend to exhibit **prime irreducibility**.

# 1. Introduction

## 1.1 Motivation

The WLM framework suggests that physical coupling constants arise from topological compatibility conditions across layers of different dimensions. The electromagnetic constant  $\alpha^{-1} = 137$  was previously derived from a triple congruence system involving:

- a 5-dimensional fiber bundle,
- a 15-dimensional normal bundle,
- a twisted Dirac index on a 12-manifold.

These congruences were computed in **cohomology**. However, cohomology only captures **additive** information. The deeper multiplicative and prime-localized structure is invisible.

## 1.2 Goal of this paper

We lift the entire dimensional-closure mechanism from cohomology to **topological K-theory and KO-theory**, and show:

1. **Gauge couplings are residues of K-theory indices localized at finite sets of primes.**
2. **Gravity corresponds to a KO-class whose localization is nontrivial at all primes.**
3. **Prime residues (e.g., 137) arise as irreducible blocks that cannot be decomposed further.**

## 1.3 Main claim

Gauge couplings = finite prime localizations of a global KO-theoretic gravitational class.

# 2. Background: K-Theory, KO-Theory, and Prime Localizations

This section develops the algebraic-topological machinery underlying the prime-localized structure of gauge couplings in the WLM. We review complex K-theory, real KO-theory, Adams operations, and the role of the J-homomorphism in distributing torsion across all primes. These ingredients collectively explain why dimensional-closure congruences naturally decompose into **prime-indexed blocks**, and why gravity corresponds to a **global KO-class** whose localizations remain nontrivial at every prime.

## 2.1 Complex K-Theory and Prime Decomposition

Let  $K^0(M)$  and  $K^1(M)$  denote the complex topological K-groups of a compact manifold  $M$ . These groups classify stable equivalence classes of complex vector bundles and exhibit **Bott periodicity**:

$$K^n(M) \cong K^{n+2}(M).$$

This periodicity implies that all complex K-groups can be expressed in terms of  $K^0$  and  $K^1$ , giving K-theory a rigid algebraic structure.

### Prime Localization

For any prime  $p$ , the localized K-group is defined by:

$$K^*(M)_{(p)} = K^*(M) \otimes \mathbb{Z}_{(p)},$$

where  $\mathbb{Z}_{(p)}$  is the localization of the integers at  $p$ . This operation isolates the  $p$ -primary information in K-theory, filtering out contributions from all other primes.

A fundamental structural fact is:

### K-theory decomposes as a product of its prime localizations

$$K^*(M) \cong \prod_{p \text{ prime}} K^*(M)_{(p)}.$$

Thus, K-theory is inherently **prime-graded**. This is the first indication that any topological invariant derived from K-theory — such as a twisted Dirac index — will naturally split into **prime-indexed congruence components**.

### Adams Operations

The Adams operations  $\psi^k: K^*(M) \rightarrow K^*(M)$  are ring homomorphisms satisfying:

- $\psi^k$  acts semisimply after localization at a prime  $p$ ,
- $\psi^p$  detects  $p$ -primary torsion,
- $\psi^k$  diagonalizes the K-theory of spheres and Lie groups.

These operations provide a canonical way to separate K-theory into **prime-specific eigenspaces**, reinforcing the idea that coupling constants — which arise from K-theoretic indices — must decompose into **prime blocks**.

## 2.2 Real KO-Theory and Gravitational Classes

While complex K-theory captures the behavior of gauge bundles, **real KO-theory** governs the topology of the tangent bundle and therefore the geometry of gravity.

Let:

$$\mathcal{G} = [TM] \in KO^{-n}(M)$$

denote the KO-class of the stable tangent bundle of an  $n$ -dimensional manifold  $M$ .  
KO-theory exhibits **8-fold Bott periodicity**:

$$KO^n(M) \cong KO^{n+8}(M).$$

### Prime Structure of KO-Theory

A crucial difference from complex K-theory is:

**KO-theory contains torsion at every prime.**

This is a deep theorem, following from:

- the structure of real vector bundles,
- the image of the J-homomorphism,
- the torsion in stable homotopy groups of spheres.

Consequently:

$$KO^*(M)_{(p)} \neq 0 \text{ for every prime } p.$$

This is the mathematical reason gravity is **global**:

- it cannot be localized to a finite prime set,
- it cannot be expressed as a finite congruence,
- it cannot produce a residue like 137.

Gravity corresponds to the **full, unlocalized KO-class**, with nontrivial components at all primes.

### 2.3 Stable Homotopy and the J-Homomorphism

The stable homotopy groups of spheres  $\pi_k^S$  contain torsion at every prime. This is one of the deepest facts in algebraic topology.

The **J-homomorphism**:

$$J: \pi_k(SO) \rightarrow \pi_k^S$$

injects geometric information from the special orthogonal group into stable homotopy. Adams proved that the image of  $J$  contains torsion at **all primes**.

### Consequences for KO-Theory

Because KO-theory receives information from  $\pi_k^S$  via the Atiyah–Bott–Shapiro construction, the torsion in KO-theory is also **prime-universal**.

Thus:

**The gravitational KO-class inherits torsion from all primes simultaneously.**

This is the deeper reason why:

- gravity = global
- gauge forces = finite prime projections
- coupling constants = residues of localized indices

and why **prime irreducibility** is a structural feature of physical couplings.

## 2.4 Summary of Section 2

We have established three key facts:

1. **K-theory decomposes into prime localizations**

$$K^*(M) \cong \prod_p K^*(M)_{(p)}.$$

2. **KO-theory is nontrivial at all primes**

$$KO^*(M)_{(p)} \neq 0 \forall p.$$

3. **The J-homomorphism distributes torsion across all primes**, making gravitational data inherently global.

These results form the algebraic-topological foundation for the WLM's interpretation of:

- gauge couplings as **finite prime residues**,
- gravity as the **global KO-closure condition**,
- and 137 as a **prime-irreducible block**.

## 3. Gauge Forces as Finite Prime Localizations

This section explains how gauge interactions arise as **finite prime localizations** of the global KO-theoretic gravitational class. Localizing the KO class at a prime  $p$  isolates the  $p$ -primary component of the geometry, and gauge bundles act as **prime filters** through their characteristic classes, dimensional gaps, and twisted Dirac indices. Only the primes dividing the characteristic numbers of the gauge bundle survive the localization, producing a finite prime set for each interaction. The localized index then yields the residue system whose Chinese-remainder intersection gives the coupling constant.

### 3.1 Localizing the gravitational KO-class

Given the global KO-class:

$$\mathcal{G} = [TM] \in KO^{-n}(M)$$

we consider its localization:

$$\mathcal{G}_{(p)} = \mathcal{G} \otimes \mathbb{Z}_{(p)}$$

### 3.2 Gauge bundles as “prime-filtered” projections

A gauge bundle  $E(U(1), SU(2), SU(3))$  interacts with  $\mathcal{G}$  through:

- twisted Dirac indices
- characteristic classes
- dimensional gaps

The key mechanism:

$\text{Index}(D \otimes E)_{(p)}$  depends only on primes dividing the characteristic numbers of  $E$ .

Thus:

- $U(1) \rightarrow \text{primes } \{2,3,5\}$
- $SU(3) \rightarrow \text{primes } \{2,3,5,7\}$
- $SU(2) \rightarrow \text{primes } \{2\}$

### 3.3 Coupling constants as residues

The localized index produces congruences:

$$k \equiv r_i \pmod{n_i}$$

where each  $n_i$  is built from **prime powers**.

The final coupling constant is the **Chinese remainder residue** of these prime blocks.

## 4. Prime Irreducibility and the Case of 137

The electromagnetic residue 137 obtained in the dimensional-closure computation is not merely a numerical coincidence. In this section we show that 137 is the **prime-irreducible residue** of the U(1) localization system. This irreducibility is a structural consequence of the prime-graded nature of K-theory, the behavior of localized indices, and the algebraic decomposition of the modulus 300. We formalize the notion of a **prime block**, prove that the U(1) congruence system decomposes into such blocks, and show that 137 is the unique integer compatible with all of them.

### 4.1 Prime Blocks and Modular Decomposition

Let  $n$  be a positive integer with prime factorization:

$$n = \prod_{i=1}^k p_i^{e_i}.$$

By the Chinese Remainder Theorem (CRT), the ring  $\mathbb{Z}/n\mathbb{Z}$  decomposes as:

$$\mathbb{Z}/n\mathbb{Z} \cong \prod_{i=1}^k \mathbb{Z}/p_i^{e_i}\mathbb{Z}.$$

Each factor  $\mathbb{Z}/p_i^{e_i}\mathbb{Z}$  is called a **prime block**. A congruence

$$x \equiv r \pmod{n}$$

is therefore equivalent to a system of congruences:

$$x \equiv r_i \pmod{p_i^{e_i}}.$$

#### Definition 4.1 (Prime-Irreducible Residue).

A residue  $r \in \mathbb{Z}/n\mathbb{Z}$  is **prime-irreducible** if its projections  $r_i$  to each prime block  $\mathbb{Z}/p_i^{e_i}\mathbb{Z}$  are individually non-decomposable and jointly compatible only for a unique integer  $r$  modulo  $n$ .

In other words, a prime-irreducible residue is the **unique intersection** of its prime blocks.

This definition is natural in the context of K-theory, where:

- each prime block corresponds to a **localized index**,

- each localized index corresponds to a **prime-specific geometric constraint**,
- the final coupling constant is the **CRT intersection** of these constraints.

## 4.2 The U(1) Modulus and Its Prime Structure

In the electromagnetic case, the dimensional-closure mechanism produced the congruences:

$$c_1(P) \equiv 1(\text{mod}5), c_1(P) \equiv -1(\text{mod}15), c_1(P) \equiv 1(\text{mod}20).$$

The combined modulus is:

$$\text{lcm}(5,15,20) = 300 = 2^2 \cdot 3 \cdot 5^2.$$

Thus the relevant prime blocks are:

- $p = 2$ : modulus 4
- $p = 3$ : modulus 3
- $p = 5$ : modulus 25

The congruences decompose as:

- mod 4:  $c_1(P) \equiv 1$
- mod 3:  $c_1(P) \equiv 2$
- mod 25:  $c_1(P) \equiv 12$

Each of these arises from a **different geometric mechanism**:

| Prime | Block  | Geometric Source                                    |
|-------|--------|-----------------------------------------------------|
| 2     | mod 4  | Spin/KO-parity of the normal bundle                 |
| 3     | mod 3  | A-genus reduction in the index theorem              |
| 5     | mod 25 | Euler class divisibility of the 5-dimensional fiber |

Thus the U(1) congruence system is already **prime-graded** before any algebraic manipulation.

## 4.3 Solving the Prime Blocks

We now solve the system:

$$\begin{cases} x \equiv 1(\text{mod}4), \\ x \equiv 2(\text{mod}3), \\ x \equiv 12(\text{mod}25). \end{cases}$$

### Step 1: Combine mod 4 and mod 3

The combined modulus is 12. Solving:

- $x \equiv 1(\text{mod}4)$  gives  $x = 1, 5, 9 \text{ mod } 12$
- Only  $x = 5$  satisfies  $x \equiv 2(\text{mod}3)$

Thus:

$$x \equiv 5(\text{mod}12).$$

### Step 2: Combine with mod 25

Solve:

$$x \equiv 5(\text{mod}12), x \equiv 12(\text{mod}25).$$

Write  $x = 12 + 25k$ . Then:

$$12 + 25k \equiv 5(\text{mod}12)$$

Since  $25 \equiv 1(\text{mod}12)$ :

$$12 + k \equiv 5(\text{mod}12) \Rightarrow k \equiv 5(\text{mod}12).$$

Thus:

$$k = 5 + 12m.$$

Substitute:

$$x = 12 + 25(5 + 12m) = 12 + 125 + 300m = 137 + 300m.$$

### Conclusion

$$\boxed{x \equiv 137(\text{mod}300)}.$$

The smallest positive solution is:

$$\boxed{137}.$$

## 4.4 Theorem: 137 Is the Prime-Irreducible Residue of the U(1) System

### Theorem 4.2.

Let  $n = 300 = 2^2 \cdot 3 \cdot 5^2$ . Let  $r \in \mathbb{Z}/n\mathbb{Z}$  satisfy the U(1) prime-block congruences:

$$r \equiv 1(\text{mod}4), r \equiv 2(\text{mod}3), r \equiv 12(\text{mod}25).$$

Then:

1. The system has a unique solution modulo  $n$ .
2. The solution is  $r \equiv 137 \pmod{300}$ .
3. The residue 137 is **prime-irreducible**: it cannot be decomposed into smaller residues under any prime localization.

**Proof.**

1. **Uniqueness** follows from the Chinese Remainder Theorem, since the moduli 4, 3, 25 are pairwise coprime.
2. **Explicit computation** (Section 4.3) gives  $r = 137$ .
3. **Prime irreducibility**:
  - The projections of 137 to the prime blocks are:
$$137 \equiv 1 \pmod{4}, 137 \equiv 2 \pmod{3}, 137 \equiv 12 \pmod{25}.$$
  - Each block is non-decomposable in its respective  $\mathbb{Z}/p^k\mathbb{Z}$ .
  - No smaller integer satisfies all three congruences simultaneously.
  - Adams operations preserve these residues, so they are stable under K-theoretic symmetries.

Thus 137 is the unique prime-irreducible residue of the  $U(1)$  localization system. ■

#### 4.5 Structural Interpretation

The irreducibility of 137 is not a physical accident but a **topological necessity**:

- Each prime block corresponds to a **localized index** of the twisted Dirac operator.
- Each localized index corresponds to a **prime-specific geometric constraint**.
- The final coupling constant is the **CRT intersection** of these constraints.
- The intersection is **prime-irreducible** because the blocks are independent.

Thus:

**137 is the smallest integer that simultaneously satisfies all prime-localized geometric constraints of the  $U(1)$  bundle.**

This explains why coupling constants tend to appear as **prime residues**: they are the irreducible building blocks of the K-theoretic index.

## 5. Gravity as the Global KO-Theoretic Closure Condition

In the WLM, gauge interactions arise from **finite prime localizations** of topological indices. Gravity, however, behaves fundamentally differently: it does not admit a finite congruence description, nor does it produce a residue analogous to 137. Instead, gravity corresponds to a **global KO-theoretic class** whose prime-localized components remain nontrivial at **every prime**. This section formalizes this distinction and shows that gravity is the **global closure condition** from which all gauge couplings arise as finite projections.

### 5.1 The KO-Class of the Tangent Bundle

Let  $M$  be an  $n$ -dimensional manifold representing a layer in the WLM. The stable tangent bundle defines a canonical KO-theory class:

$$\mathcal{G} = [TM] \in KO^{-n}(M).$$

This class encodes the global geometric structure of the layer:

- orientation,
- spin structure,
- Pontryagin classes,
- the behavior of the Dirac operator,
- and the interaction of the manifold with stable homotopy.

Unlike gauge bundles, which live naturally in **complex K-theory**, the tangent bundle is inherently a **real** object, and therefore belongs to **KO-theory**.

#### KO-Theory Is Prime-Universal

A central fact is:

**KO-theory contains torsion at every prime.**

This follows from:

- the structure of  $\pi_k(SO)$ ,
- the image of the J-homomorphism,
- the torsion in  $\pi_k^S$ ,
- and the Atiyah–Bott–Shapiro construction.

Thus:

$$KO^*(M)_{(p)} \neq 0 \text{ for every prime } p.$$

This is the first indication that gravity is **global**: it cannot be restricted to a finite prime set.

## 5.2 Prime Localization of the Gravitational Class

For any prime  $p$ , define the localized gravitational class:

$$\mathcal{G}_{(p)} = \mathcal{G} \otimes \mathbb{Z}_{(p)}.$$

Because KO-theory is nontrivial at all primes, we have:

$$\mathcal{G}_{(p)} \neq 0 \forall p.$$

This is in stark contrast to gauge bundles  $E$ , whose localized indices vanish for all but finitely many primes.

### Interpretation

- **Gauge forces**: arise from **finite prime subsets** where the localized index is nontrivial.
- **Gravity**: arises from the **entire prime spectrum**, with no vanishing localizations.

Thus:

**Gravity = the only interaction whose prime support is infinite.**

This is the mathematical meaning of “gravity is global.”

## 5.3 Why Gravity Cannot Produce a Residue

Gauge couplings arise from congruence systems of the form:

$$k \equiv r_i \pmod{n_i},$$

where each modulus  $n_i$  is built from **prime powers** associated with:

- dimensional gaps,
- characteristic-class divisibility,
- localized indices.

The final coupling constant is the **CRT intersection** of these prime blocks.

Gravity cannot produce such a residue for three reasons:

### (1) KO-Theory Has Infinite Prime Support

Since:

$$KO^*(M)_{(p)} \neq 0 \forall p,$$

gravity would require a congruence system with **infinitely many moduli**:

$$k \equiv r_p \pmod{p^{e_p}} \text{ for all primes } p.$$

No finite integer can satisfy infinitely many independent congruences.

## **(2) The Tangent Bundle Is Not a Gauge Bundle**

Gauge bundles are:

- complex,
- finite rank,
- associated to compact Lie groups.

The tangent bundle is:

- real,
- globally defined,
- tied to the manifold's topology,
- not a representation of a gauge group.

Thus its index behavior is fundamentally different.

## **(3) The Dirac Operator on the Tangent Bundle Produces Global Invariants**

The index of the Dirac operator twisted by the tangent bundle produces:

- the  $\hat{A}$ -genus,
- Pontryagin numbers,
- KO-valued invariants,

none of which reduce to finite congruences.

Thus:

**Gravity cannot be expressed as a finite residue because its KO-theoretic invariants are global, not modular.**

## **5.4 Dimensional Closure Revisited**

In the  $U(1)$  case, dimensional gaps (5, 15, 20) acted as **prime filters**:

- gap 5  $\rightarrow$  prime 5
- gap 15  $\rightarrow$  primes 3 and 5
- gap 20  $\rightarrow$  primes 2 and 5

These filters produced the moduli:

$$5, 15, 20,$$

whose least common multiple is:

$$300 = 2^2 \cdot 3 \cdot 5^2.$$

Gravity behaves differently:

**Dimensional gaps do not filter gravity.**

Because the tangent bundle is present in **every dimension**, and KO-theory is nontrivial at **every prime**, dimensional gaps cannot restrict gravity to a finite prime set.

Thus:

**Dimensional closure for gravity is global, not modular.**

This is why gravity does not produce a number like 137.

### 5.5 Gravity as the Global Closure Condition

We now formalize the central claim.

**Theorem 5.1 (Gravity Is the Global KO-Closure Condition).**

Let  $\mathcal{G} = [TM] \in KO^{-n}(M)$  be the gravitational KO-class. Then:

1.  $\mathcal{G}_{(p)} \neq 0$  for all primes  $p$ .
2. No finite set of primes  $S$  satisfies  $\mathcal{G}_{(p)} = 0$  for  $p \notin S$ .
3. Therefore gravity cannot be expressed as a finite congruence or residue.
4. Gauge couplings arise from finite prime localizations of  $\mathcal{G}$ .
5. Gravity is the **global closure condition** from which all gauge couplings descend.

**Proof.**

1. Nontriviality of  $\mathcal{G}_{(p)}$  follows from the prime-universal torsion in KO-theory.
2. If such a finite set  $S$  existed, KO-theory would vanish outside  $S$ , contradicting the structure of  $\pi_k^S$ .
3. A finite residue requires a finite modulus, which requires a finite prime set; gravity has infinite prime support.
4. Gauge bundles  $E$  restrict the index to finitely many primes, producing residues.
5. Thus gravity is the global object whose finite projections yield gauge couplings. ■

### 5.6 Structural Interpretation

The distinction between gravity and gauge forces is now clear:

| Interaction  | Prime Support | Mathematical Object               | Output                      |
|--------------|---------------|-----------------------------------|-----------------------------|
| Gravity      | All primes    | KO-class of tangent bundle        | Global invariant            |
| Gauge forces | Finite primes | K-theory classes of gauge bundles | Residue (coupling constant) |

Thus:

**Gravity is the global KO-closure condition. Gauge forces are finite prime localizations. Coupling constants are residues of these localizations.**

This is the structural heart of the WLM.

## 6. A Unified Framework for Physical Couplings in the WLM

The preceding sections have developed the algebraic, geometric, and homotopical machinery underlying the WLM’s interpretation of physical coupling constants. We now synthesize these ingredients into a unified topological framework. The central idea is that **all physical couplings arise from the same global KO-theoretic gravitational class**, but appear numerically as **finite prime residues** when projected through the geometric constraints of gauge bundles and dimensional gaps.

This section formalizes this framework and shows how the WLM organizes physical interactions into a coherent hierarchy:

- **gravity** = global KO-class (all primes active)
- **gauge forces** = finite prime localizations
- **coupling constants** = residues of localized indices

This provides a single structural explanation for the appearance of prime irreducibility in physical constants such as 137.

### 6.1 Dimensional Gaps Determine the Moduli

In the WLM, each interaction is associated with a sequence of embeddings:

$$M_a \hookrightarrow M_b \hookrightarrow M_c,$$

where the **dimensional gaps**:

$$\Delta_1 = \dim M_b - \dim M_a, \Delta_2 = \dim M_c - \dim M_b,$$

determine the **moduli** of the congruence system.

This arises because:

- Euler classes depend on the rank of the normal bundle,
- Pontryagin classes depend on the dimension of the tangent bundle,
- Gysin maps shift degrees by the dimension of the fiber,
- index formulas involve characteristic numbers whose divisibility depends on dimension.

Thus:

**Dimensional gaps act as geometric filters that select which primes appear in the modulus.**

For example:

- gap 5  $\rightarrow$  modulus divisible by 5
- gap 15  $\rightarrow$  modulus divisible by 3 and 5
- gap 20  $\rightarrow$  modulus divisible by 2 and 5

The combined modulus is the least common multiple of these contributions.

## 6.2 Characteristic Classes Determine Prime Factors

Each characteristic class has intrinsic divisibility properties tied to specific primes:

- Euler class of odd rank  $\rightarrow$  mod 2
- Pontryagin classes  $\rightarrow$  mod 3, mod 5
- $c_2$  of  $SU(3)$   $\rightarrow$  primes 2,3,5,7
- $\hat{A}$ -genus  $\rightarrow$  mod 3
- Chern character  $\rightarrow$  rational but with prime-specific denominators

Thus:

**Characteristic classes determine which primes survive the dimensional filters.**

The modulus  $n$  of the congruence system is therefore:

$$n = \prod_{p \in S} p^{e_p},$$

where  $S$  is the finite set of primes selected by:

- the dimensional gaps, and
- the characteristic-class divisibility of the gauge bundle.

### 6.3 Localized Indices Determine the Residues

Given a gauge bundle  $E$ , the twisted Dirac index:

$$\text{Index}(D \otimes E)$$

encodes the interaction between:

- the geometry of the manifold,
- the topology of the gauge bundle,
- and the global KO-class of the tangent bundle.

After localization at a prime  $p$ :

$$\text{Index}(D \otimes E)_{(p)} \in \mathbb{Z}_{(p)},$$

the index reduces to a **prime-specific residue**:

$$k \equiv r_p \pmod{p^{e_p}}.$$

Each residue  $r_p$  is determined by:

- the parity of the Dirac operator (mod 2),
- the  $\hat{A}$ -genus (mod 3),
- the Euler class of the fiber (mod 5),
- higher Pontryagin numbers (mod 7, mod 11, ...).

Thus:

**The residue is the prime-localized shadow of the global KO-class.**

The final coupling constant is the **CRT intersection** of these residues.

### 6.4 Gauge Forces as Finite Prime Localizations

A gauge bundle  $E$  interacts with the gravitational KO-class  $\mathcal{G}$  through the index:

$$\text{Index}(D \otimes E).$$

However, this interaction is **prime-selective**:

- only primes dividing the characteristic numbers of  $E$  contribute,
- only primes compatible with the dimensional gaps survive,
- all other primes vanish under localization.

Thus:

$$\text{Index}(D \otimes E)_{(p)} = 0 \text{ for all but finitely many primes } p.$$

This yields:

**Gauge forces = finite prime localizations of the global KO-class.**

Examples:

- $U(1) \rightarrow \text{primes } \{2, 3, 5\} \rightarrow \text{residue } 137$
- $SU(2) \rightarrow \text{prime } \{2\}$
- $SU(3) \rightarrow \text{primes } \{2, 3, 5, 7\}$  (to be computed in Paper B)

## 6.5 Gravity as the Global KO-Closure Condition

In contrast, the gravitational class:

$$\mathcal{G} = [TM] \in KO^{-n}(M)$$

has **nontrivial localization at every prime**:

$$\mathcal{G}_{(p)} \neq 0 \forall p.$$

Thus:

- gravity cannot be expressed as a finite congruence,
- gravity cannot produce a residue,
- gravity is the **global object** from which all gauge couplings descend.

This yields the structural hierarchy:

| Level            | Object             | Prime Support  | Output              |
|------------------|--------------------|----------------|---------------------|
| <b>Global</b>    | Gravity (KO-class) | All primes     | Global invariant    |
| <b>Local</b>     | Gauge forces       | Finite primes  | Coupling constant   |
| <b>Numerical</b> | Coupling constant  | Prime residues | Integer (e.g., 137) |

## 6.6 Master Equation

All physical couplings in the WLM satisfy:

|                                                                                                     |
|-----------------------------------------------------------------------------------------------------|
| $\text{Coupling constant} = \text{Index}(D \otimes E) \text{ localized at a finite set of primes.}$ |
|-----------------------------------------------------------------------------------------------------|

This single equation unifies:

- dimensional gaps  $\rightarrow$  moduli
- characteristic classes  $\rightarrow$  prime factors
- index theory  $\rightarrow$  residues
- K-theory  $\rightarrow$  prime localization
- KO-theory  $\rightarrow$  globality of gravity

into one coherent topological mechanism.

## 7. Conclusion and Outlook

In this paper we have developed a K-theoretic refinement of the dimensional-closure mechanism in the World-Layer Model (WLM), revealing a unified topological structure underlying physical coupling constants. The key insight is that all couplings arise from **prime-localized projections** of a single **global KO-theoretic gravitational class**, and that numerical residues such as 137 correspond to **prime-irreducible blocks** that survive dimensional projection.

This perspective reframes the role of coupling constants in fundamental physics. Rather than being arbitrary empirical parameters, they emerge as **topological residues** of index-theoretic invariants, filtered through the geometry of dimensional gaps and the divisibility properties of characteristic classes. The WLM thus provides a structural explanation for the appearance of prime numbers and prime-power moduli in physical constants.

### 7.1 Summary of Key Results

We summarize the main conceptual and mathematical contributions:

#### (1) Prime-graded structure of K-theory

Complex K-theory decomposes into prime localizations:

$$K^*(M) \cong \prod_p K^*(M)_{(p)},$$

and Adams operations separate these components. This explains why coupling constants naturally decompose into **prime blocks**.

#### (2) KO-theory is nontrivial at all primes

The gravitational KO-class:

$$\mathcal{G} = [TM] \in KO^{-n}(M)$$

has nonvanishing localization at **every prime**, due to the prime-universal torsion in stable homotopy and the image of the J-homomorphism. This establishes gravity as the **global closure condition**.

#### (3) Gauge forces as finite prime localizations

Gauge bundles  $E$  interact with  $\mathcal{G}$  through the twisted Dirac index. Only finitely many primes survive the combined filters of:

- dimensional gaps,

- characteristic-class divisibility,
- and localized index behavior.

Thus:

$$\text{Index}(D \otimes E)_{(p)} = 0 \text{ for all but finitely many primes } p.$$

Gauge forces are therefore **finite prime projections** of the global KO-class.

#### (4) Coupling constants as prime residues

The final coupling constant is the **Chinese-remainder intersection** of the prime-localized residues:

$$k = \bigcap_{p \in S} r_p.$$

For  $U(1)$ , this yields the prime-irreducible residue:

$$137 \equiv 1 \pmod{4}, 137 \equiv 2 \pmod{3}, 137 \equiv 12 \pmod{25}.$$

This demonstrates that 137 is not accidental but structurally enforced.

## 7.2 Conceptual Implications

The WLM's unified framework suggests a new interpretation of physical constants:

- **They are not arbitrary.** They arise from topological compatibility conditions across layers.
- **They are not continuous parameters.** They are discrete residues of index-theoretic invariants.
- **They are not independent.** They are finite projections of the same global KO-class.
- **They are prime-structured.** Their irreducibility reflects the prime decomposition of K-theory.

This provides a coherent explanation for the long-observed appearance of prime numbers in physical constants.

## 7.3 Outlook: Toward $SU(3)$ and Beyond

The next step is to apply the same machinery to non-abelian gauge groups.

**$SU(3)$**

The  $SU(3)$  bundle introduces new characteristic classes  $(c_2, p_1)$  and higher-degree terms in the index formula. These bring in additional primes:

$$\{2, 3, 5, 7\},$$

and possibly 11 through stable homotopy. The resulting congruence system will yield the **strong-interaction coupling constant** as a new prime-irreducible residue.

## **SU(2)**

The  $SU(2)$  case is simpler, dominated by the prime 2, and provides a clean intermediate test.

## **Gravity**

Further work will formalize the KO-globality of gravity and its role as the universal closure condition.

## **7.4 Final Perspective**

The WLM reveals a strikingly simple and elegant structure:

$$\boxed{\text{Coupling constants} = \text{Index}(D \otimes E) \text{ localized at a finite set of primes.}}$$

Gravity provides the **global KO-class**. Gauge forces are its **finite prime shadows**. Coupling constants are the **residues** of these shadows.

This unifies the topology of vector bundles, the arithmetic of primes, and the numerical structure of physical constants into a single coherent framework.

## **Appendix A. K-theory localization machinery**

## **Appendix B. Adams operations and prime torsion**

## **Appendix C. Relation to stable homotopy groups of spheres**

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